

PRACTICAL 11:

AIM:

Design Web Applications using Server-Side Session Management

STEP 1: CREATE PROJECT

1. Open Visual Studio
2. Click **Create New Project**
3. Select:
ASP.NET Web Application (.NET Framework)
4. Click Next
5. Select:
Web Forms
6. Enter Project Name: **ServerStateDemo**
7. Click Create

STEP 2: ADD WEB FORMS

1. Right click Project → Add → Web Form
2. Name:
 1. **Home.aspx**
 2. **Login.aspx**

Step 3: Design Login Page (Login.aspx)

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Login.aspx.cs"
Inherits="ServerStateDemo.Login" %>
```

```
<!DOCTYPE html>
```

```
<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>
```

```

Username:
<asp:TextBox ID="txtUser" runat="server"></asp:TextBox>

<br /><br />

Password:
<asp:TextBox ID="txtPass" runat="server" TextMode="Password"></asp:TextBox>

<br /><br />

<asp:Button ID="btnLogin" runat="server" Text="Login" OnClick="btnLogin_Click" />

    </div>
</form>
</body>
</html>

```

Step 4: Login Code (Login.aspx.cs)

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace ServerStateDemo
{
    public partial class Login : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {

        }

        protected void btnLogin_Click(object sender, EventArgs e)
        {
            if (txtUser.Text == "admin" && txtPass.Text == "123")
            {
                // SESSION (per user)
                Session["username"] = txtUser.Text;

                // APPLICATION (global count)
                Application.Lock();
                Application["users"] = Convert.ToInt32(Application["users"]) + 1;
                Application.Unlock();

                Response.Redirect("Home.aspx");
            }
            else
            {
                Response.Write("Invalid Login");
            }
        }
    }
}

```

Step 5: Design Home Page (Home.aspx)

```

<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Home.aspx.cs"
Inherits="ServerStateDemo.Home" %>

<!DOCTYPE html>

<html xmlns="http://www.w3.org/1999/xhtml">
<head runat="server">
    <title></title>
</head>
<body>
    <form id="form1" runat="server">
        <div>

            Welcome,
<asp:Label ID="lblUser" runat="server"></asp:Label>

<br /><br />

Total Users:
<asp:Label ID="lblUsers" runat="server"></asp:Label>

<br /><br />

Total Hits:
<asp:Label ID="lblHits" runat="server"></asp:Label>

<br /><br />

<asp:Button ID="btnLogout" runat="server" Text="Logout" OnClick="btnLogout_Click" />

        </div>
    </form>
</body>
</html>

```

Step 6: Home Code (Home.aspx.cs)

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace ServerStateDemo
{
    public partial class Home : System.Web.UI.Page
    {
        protected void Page_Load(object sender, EventArgs e)
        {
            if (Session["username"] != null)
            {
                // SESSION DATA
                lblUser.Text = Session["username"].ToString();

                // APPLICATION DATA
                lblUsers.Text = Application["users"].ToString();
                lblHits.Text = Application["hits"].ToString();
            }
            else
            {

```

```

        Response.Redirect("Login.aspx");
    }

    // Count hits
    Application.Lock();
    Application["hits"] = Convert.ToInt32(Application["hits"]) + 1;
    Application.Unlock();
}

protected void btnLogout_Click(object sender, EventArgs e)
{
    Session.Abandon();
    Response.Redirect("Login.aspx");
}
}
}

```

Step 7: Global.asax

```

using System;
using System.Collections.Generic;
using System.Linq;
using System.Web;
using System.Web.Security;
using System.Web.SessionState;

namespace ServerStateDemo
{
    public class Global : System.Web.HttpApplication
    {
        protected void Application_Start(object sender, EventArgs e)
        {
            Application["users"] = 0;
            Application["hits"] = 0;
        }
    }
}

```

Output:

Username:

Password:

Welcome, admin

Total Users: 1

Total Hits: 0

Logout

Welcome, admin

Total Users: 1

Total Hits: 1

Logout